Week 1 Notes & Quiz

Notes for all five Week 1 lectures, made only from the 127 slides in *WEEK 1.pdf*. Each yellow tag such as S27 is the slide (PDF page) the point comes from. The 50-question practice questions and answer key are at the end.

STOP	LECTURE	SLIDES
L1	**Introduction to Logistics: Design and Management**	S1–S13
L2	**Logistics Stakeholders and Modern Logistics Concepts**	S14–S32
L3	**Manufacturing Logistics: National and International Scenario**	S33–S65
L4	**Storage and Warehousing**	S66–S89
L5	**Logistics Services Providers**	S90–S127
+	**Cheat sheet · Slide quirks · Quiz (50 Qs)**	—

LECTURE 01 S1–S13

Introduction to Logistics: Design and Management

Covers: definition, logistics evolution, key elements of logistics management, conclusion, books S2

What logistics is

> DEFINITION · EXACT WORDS
>
> *"Logistics refers to the management of the flow of goods, information, and resources from the point of origin to the point of consumption."* S7

- The flow is forward as well as reverse. S7
- It covers planning, implementing and controlling the efficient movement and storage of goods, plus the related information and financial transactions, within a supply chain. S7
- Hook used in class: your latest online purchase arrives with one click; logistics is the "behind-the-scenes magic" that gets it there. S6

The Seven R's 7 items

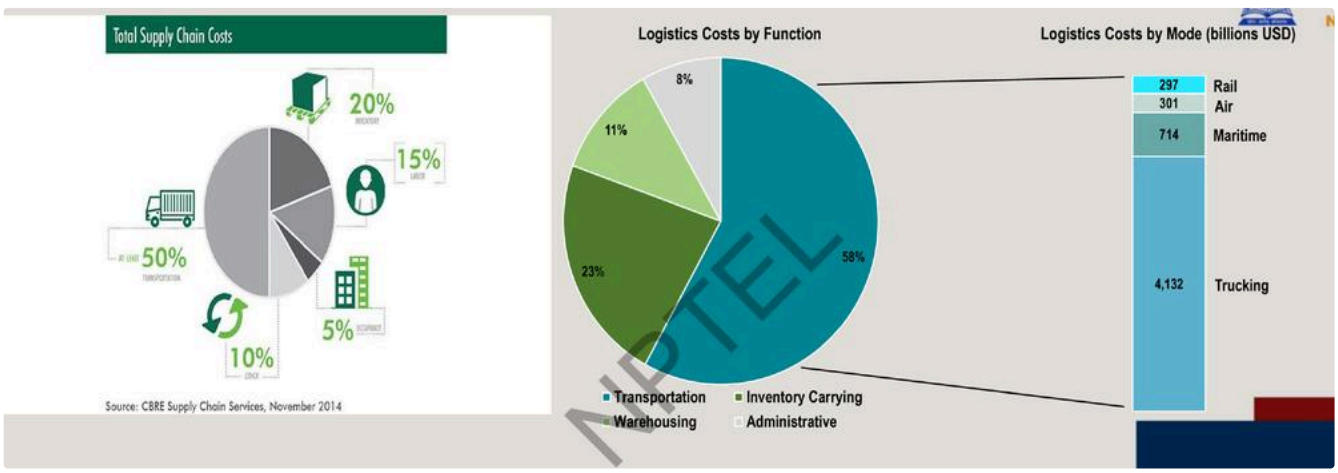
Best practices in logistics management S8:

1. Right **product** (including the right information about it)
2. (At) right **quantity**
3. Right **time**
4. Right **condition**
5. Right **place**
6. (to) the right **customer**
7. (with the) right (financial) **resources**

Logistics evolution

`Local` → `Regional` → `National` → `Global` S5

Where the money goes



Original slide S3. Left: Total Supply Chain Costs (CBRE, Nov 2014). Middle: Logistics Costs by Function. Right: Logistics Costs by Mode, billions USD.

CHART	VALUES ON THE SLIDE
Total supply chain costs	Transportation **at least 50%** · Inventory 20% · Labor 15% · Other 10% · Occupancy 5%
Costs by function	Transportation **58%** · Inventory carrying 23% · Warehousing 11% · Administrative 8%
Costs by mode (bn USD)	Trucking **4,132** · Maritime 714 · Air 301 · Rail 297

- Two video slides (a 1998 Sydney clip, and 10-minute delivery shorts) ask: "With the pressure to deliver, what if something goes wrong?" Only links are in the PDF. S4 S9

Key elements of logistics 5 items

Transportation · warehousing · inventory control · order fulfillment · distribution S10

> ULTIMATE AIM · EXACT WORDS
>
> *"...to deliver the correct products to the right location at the appropriate time, all while minimizing expenses and maximizing effectiveness."* S10

- Competent logistics management helps companies meet customer needs, improve operational efficiency and keep a competitive edge. S10
- Conclusion: effective logistics optimizes operations, reduces costs, improves customer satisfaction, and is indispensable for business competitive advantage. S11

Text and reference books

1. Donald Waters, *Logistics: An Introduction to Supply Chain Management* (Palgrave)
2. Ghiani, Laporte & Musmanno, *Introduction to Logistics Systems Management* (Wiley)
3. John J. Liu, *Supply Chain Management and Transport Logistics* (Routledge)
4. Mangan & Lalwani, *Global Logistics and Supply Chain Management* (Wiley)
5. Bramel & Simchi-Levi, *The Logic of Logistics*

Listed on S12, and again on S31 S64 S126

LECTURE 02 S14–S32

Logistics Stakeholders and Modern Logistics Concepts

Covers: stakeholders overview, advantages of effective logistics management, logistics design and management, modern concepts S15

Stakeholders and elements

The diagram puts **Logistics** at the centre of: Customers, Suppliers, Procurement, Manufacturers, Products, Inventory, Retail, Transport and Distribution. S16

The process, step by step S17 S18:

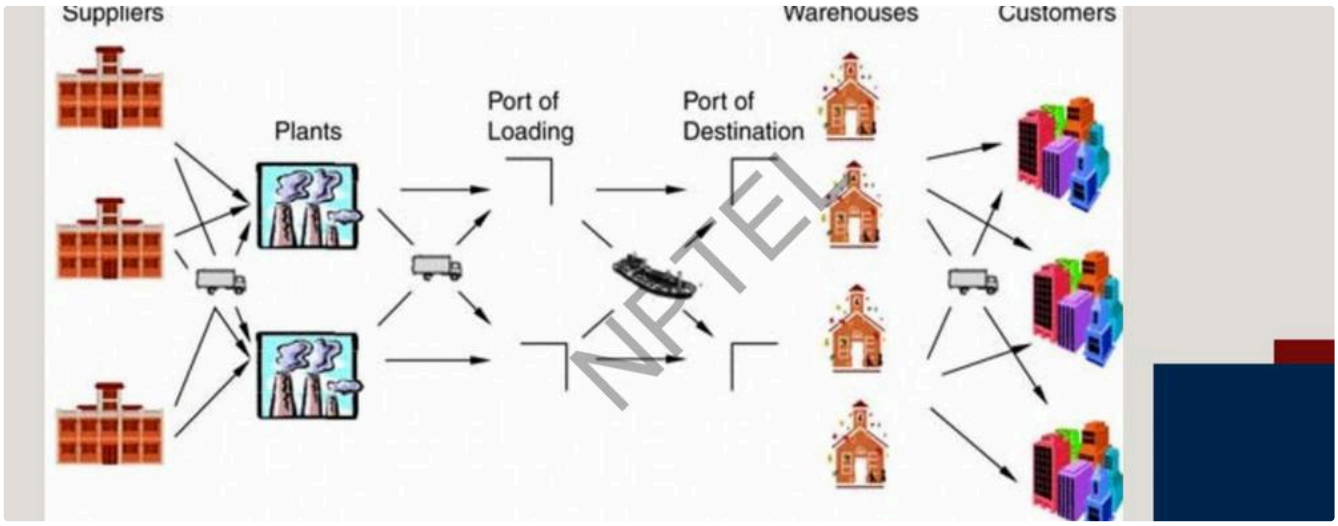
STEP	WHAT HAPPENS
Order placement	Customers place orders for goods or services.
Order processing	Done by the company's sales or customer service team.
Inventory management	Stock levels checked regularly to avoid stock-outs; order cycles continue to replenish raw-material stock.
Warehousing	Products stored in warehouses at strategic locations.
Picking & packing	Orders picked from shelves and packed securely before shipping.
Transportation	Packed orders move from warehouse to destination by truck, train, ship or airplane.
Delivery	To the customer's location; local carriers or postal services can do the last mile.
Order tracking	Customers track orders with docket numbers given by the logistics service provider.
Customer service	Handles inquiries, issues and after-sale returns.
Feedback & improvement	Customer feedback and internal performance metrics show where to improve.

> LOGISTICS MANAGEMENT · EXACT WORDS
>
> *"Logistics management involves the planning, implementation, and control of the movement and storage of goods, services, and related information within a supply chain."* S19

Advantages of effective logistics management 7 items

Efficiency	Seamless movement of goods, services and information; fewer interruptions. S20
Cost reduction	Optimized routes, warehouse operations and inventory levels cut storage, transport and holding costs. S20
Customer satisfaction	Timely delivery builds satisfaction and long-term loyalty. S20
Competitive advantage	Better service, faster delivery, lower cost. S20
Risk management	Mitigates stock-outs, transport delays and disruptions; ensures business continuity. S21
Resource optimization	Best use of vehicles, warehouses and manpower. S21
Market expansion	Reach new customers in different geographic regions efficiently. S21

Logistics design and management



Original slide S22, "Logistics Designing and Management" (source: thebrimichgroup.com).

`Suppliers` → `Plants` → `Port of Loading` → `Port of Destination` → `Warehouses` → `Customers`

- Strategic planning, coordination and execution of supply chain processes for an efficient flow of goods, information and resources from origin to consumption. S23
- Includes inventory management, warehousing, transportation, order processing and distribution. S23
- Good design weighs **cost optimization, customer satisfaction, sustainability and risk management**. S23
- Involves analysing demand patterns, choosing transport modes, designing warehouse layouts, inventory control systems, and real-time tracking. It needs continuous evaluation and improvement. S23

Core elements 8 items

Strategic planning	Define logistics objectives, analyse market demand, align with business goals. S24
Inventory management	Balance supply and demand through forecasting, optimization and cost-effective stock management. S24
Warehousing	Storage efficiency, effective material handling, inventory control. S24
Transportation	Right modes, route optimization, freight management. S24
Order processing	Timely, accurate processing from entry to delivery. S25
Distribution	Timely delivery, last-mile optimization, cost minimization. S25
Technology incorporation	RFID, IoT, AI-ML, UAV and GPS for real-time monitoring and optimization. S25
Continuous improvement	Analyse KPIs, find areas to improve (reachability, deliverability), adapt to market changes. S25

Modern logistics concepts 8 items

Innovative strategies that optimize supply chain efficiency, improve customer satisfaction and build competitive advantage. S26

CONCEPT	CORE IDEA
Lean	From lean manufacturing: **waste reduction**, streamlined processes, maximum value. S27
Just-in-Time (JIT)	Minimize **inventory holding costs** by matching production and delivery to customer demand; shorter lead times. S27
Agile	**Flexibility and responsiveness** to fast-changing markets: dynamic networks, real-time analytics, close supplier collaboration. S27
Green	Also called **sustainable logistics**: better routes, eco-friendly fuels and vehicles, less packaging, recycling. S28
Collaborative	**Cooperation and partnership** among stakeholders; share information, resources, best practices. S28
Omni-channel	Driven by e-commerce and multi-channel retail; consistent experience across all sales channels. S28
Reverse	**Returns, exchanges, end-of-life disposal**; recover value, cut waste, meet recycling rules. S29
Supply chain visibility	Real-time tracking of inventory and shipments using RFID, AI-ML, IoT, UAV, GPS and cloud software. S29

Conclusion: logistics design and management is a dynamic field needing careful planning and execution so goods, information and resources flow smoothly among stakeholders. S30

LECTURE 03 S33–S65

Manufacturing Logistics: National and International Scenario

Covers: manufacturing logistics context and architecture, factors behind logistics development, current status globally and in India S34

> MANUFACTURING LOGISTICS · EXACT WORDS
>
> *"Manufacturing logistics entails overseeing the materials, resources, and procedures integral to producing goods within a manufacturing setting."* S35
>
> Tasks: **procurement, inventory control, scheduling production, managing transportation, distributing finished products**. S35

Architecture of manufacturing logistics 8 items

Supply chain management	Seamless flow of materials from suppliers to the plant, and final product to the customer. S36
Inventory management	Balance inventory: low holding cost, still meet production demand. S36
Production planning & scheduling	Meet demand while minimizing lead times and production cost. S36
Quality control	Quality of raw materials, work-in-progress and finished goods via inspection and QA. S36
Material handling	Movement of material *within* the facility with suitable equipment and processes. S37
Transportation	Between suppliers, plants, distribution centres and customers. S37
Warehousing	Storage and handling of materials and finished goods. S37
Reverse logistics	Return of defective or excess materials; recycling or disposal of waste. S37

S38 is a link to an Ekart LinkedIn post; no content in the PDF.

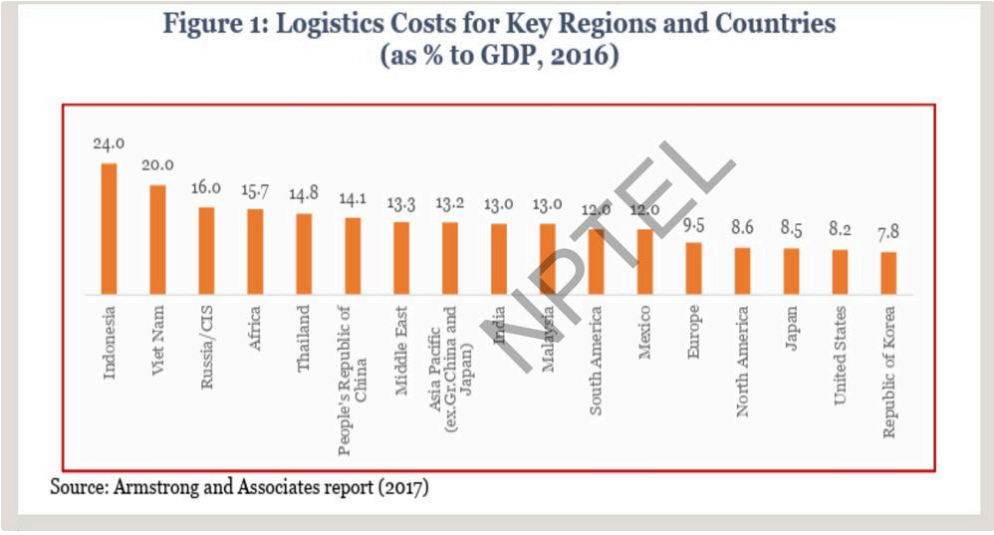
Factors behind the development of logistics 9 items

Globalization	International trade made supply chains complex; needs advanced global solutions. S39
E-commerce growth	Faster delivery, flexible fulfilment, integration across sales channels. S40
Supply chain collaboration	Alliances among suppliers, manufacturers, distributors, LSPs. S41
Advancements in technology	GPS tracking, RFID, cloud computing, data analytics. S41
Infrastructure development	Ports, airports, railways, highways; shorter transit times. S42
Emerging markets	Opportunities and challenges; new dynamics and regulations. S42
Sustainability initiatives	Alternative fuels, green packaging, energy-efficient modes. S43
Customer expectations	Faster delivery, accurate tracking, hassle-free returns. S43
Regulatory changes	Trade, customs, safety and environmental standards. S44

What is happening globally

FACT	VALUE
Industry size 2021	Over **8.4 trillion euros**; projected to exceed **13.7 trillion euros by 2027** S47
Total logistics costs 2020	**9 trillion USD ≈ 10.7% of global GDP** (GDP 85.24 trillion USD) S47
Largest share of logistics costs	**Asia-Pacific** region S47
Largest 3PL market	**United States** S47
Emerging markets ranking	Agility Emerging Markets Logistics Index S47
Market size chart (trillion USD)	2018 5.58 · 2019 6 · 2020 5.73 · 2021 8.43 · 2022 10.41 · 2023* 10.71 · 2027* 13.71 · 2028* 14.08 S45

S47 also mentions the global supply chain pressure index fluctuating, and rising goods exports with e-commerce playing a big role.



Original slide S46, Figure 1: Logistics Costs for Key Regions and Countries (as % to GDP, 2016), Armstrong & Associates (2017) via NCAER. Highest: Indonesia 24.0. Lowest: Republic of Korea 7.8. India: 13.0. USA: 8.2.

What is happening in India

FACT	VALUE
Market size 2022	About **$274 billion** S49
Projection 2030	**$563 billion** at **9.4% CAGR** S49
Logistics cost / GDP	**14%** in India vs BRICS average **11%** S49
Why so costly	Inefficient **modal mix**, especially the **road** segment; lags other BRICS in efficiency S49
Growth context	India the **2nd fastest-growing nation, next only to China**; GDP 1.3 → 2.9 USD tn (2009→2019), CAGR 8.4% (China 10.9%, World 3.8%) S48

- Logistics is the **backbone** of every sector's supply chain (healthcare, hospitality, manufacturing). S50
- Everything is time-bound: a small delay at one stage causes a magnified ripple effect; a few hours of downtime can become days of delay. S50
- Common target: cut the supply-cycle time by streamlining processes. S51

Challenges for Indian logistics companies 12 items

CHALLENGE	KEY POINT
High order intensity ratio	Bulk orders overwhelm limited resources; **festival time** is the best example of mass delays and ETA failure. S52
Transportation roadblocks	Bad interior roads, hilly accidents and landslides, congestion, checkpoints, tolls. S53
Rail tariffs	High rail costs and tariff hikes, congested network, monsoon floods cancel services. S54
Port and shipping problems	Shipping is the slowest mode; understaffed ports, overstuffed berths; coastal ports lack **depth** for large vessels. S55
Lack of skilled personnel	Under-skilled, over-worked staff; leads to high turnover and training costs. S56
Slow move to new technology	Compared with **Japan**. Costs: set-up and installation, foreign trainers, time to learn and integrate. S57
Warehousing & taxation	Pest-infested, leaky, cramped warehouses; government uses the biggest ones to store **grains**; layered taxes waste time. S58
Competition with global giants	Short cycles at low prices and free one-day delivery by global players. S59
Customer's mindset	Customers won't pay extra even for **under 2-day** delivery; Amazon-level expectations. S60
Ever-increasing fuel costs	Raise the **surcharge on freight tariffs**, cutting profit. S61
Government policies & bottlenecks	Tighter compliance mandates and regulation. S62
Shortage of drivers & delivery staff	Hard to hire and retain; remaining staff burn out. S62

LECTURE 3 CONCLUSION S63

Manufacturing logistics = procurement, inventory control, scheduling production, managing transportation, distributing final products. Major forces = globalization, e-commerce, supply chain collaboration, technology, infrastructure, emerging markets, sustainability, customer expectations, regulatory bodies.

LECTURE 04 S66–S89

Storage and Warehousing

Covers: storage, warehousing, cross docking (utilities, challenges, impacts), case studies S67

> STORAGE · EXACT WORDS
>
> *"Storage in logistics and supply chain management refers to the process of storing goods, materials, and products at various points along the supply chain, from raw materials procurement to finished goods distribution."* S68
>
> Good storage management keeps materials flowing, optimizes inventory and meets demand. S68

- Blinkit, Zepto and other 10-minute deliveries come from dark stores, which shows why storage matters. S69
- The storage diagram links five "rights" (time, price, place, products, customers) to store operations, customer, category, supplier, logistics & warehouse, and forecasting/replenishment/promotions management, around optimized business processes and an integrated operations database. S70

Components of storage

Warehousing	Storage hubs before distribution; owned by manufacturers, wholesalers, retailers or **3PLs**; shelving, racks, pallets. S71
Inventory management	Enough stock, low holding cost: optimization, replenishment, cycle counting, valuation. S71
Storage facilities	Warehouses, distribution centres, **cold storage** (perishables), **bonded warehouses** (goods under customs control), **cross-docking** facilities (immediate transfer without storage). S71
Storage equipment	Pallet racks, shelving, mezzanine floors, forklifts, conveyors, **AS/RS** (automated storage and retrieval systems). S72
Safety & security	Cameras, access control, alarms, barriers; safety protocols prevent accidents. S72
Material handling	Moving, protecting, storing and controlling materials from manufacturing to distribution. Manual (physical force) vs automated (robots, computerized devices). S72

Warehousing

> WAREHOUSE · EXACT WORDS
>
> *"A warehouse is a commercial facility designed for the storage of goods, materials, and products."* S74
>
> A centralized hub where items are stored **temporarily** before distribution; uses shelving, racks, pallets and inventory systems. S74

Warehouse layout design: FAST

F · Flow	Stock inflow and outflow strategy. S75
A · Accessibility	Enough room for staff and forklifts to move safely and quickly. S75
S · Space	Maximum space for operational storage and stock processing; minimum for offices, monitoring, empty pallet storage. S75
T · Throughput	Units processed per time interval; usually measures **outbound** operations. S75

7 steps to plan the layout S76

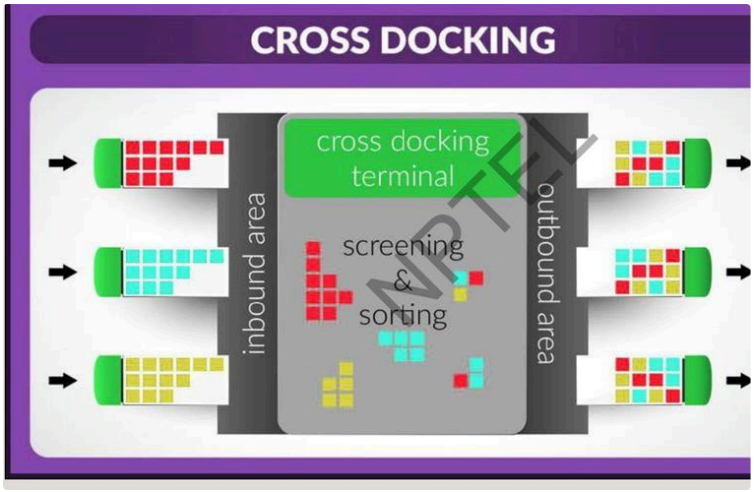
1. Define the overall purpose of the warehouse
2. Create a warehouse diagram and calculate the total storage area
3. Consider storage requirements
4. Evaluate material handling equipment needs
5. Map it out and optimize space
6. Test traffic flow
7. Implement the layout

COMMON LAYOUT	BEST FOR S76
U shape	The **most common**; works with any type of warehouse.
I shape	Through-flow; **high volumes**; separates loading and shipping areas.
L shape	Lots of **storage needs**; greater sorting areas.

Utilities of warehouses 7 items

Product consolidation	Combine goods from many suppliers into larger shipments; lower transport cost. S78
Cross-docking	Unload, sort and load straight onto outbound trucks; little or no storage. S78
Value-added services	Packaging, labelling, **kitting**, assembly. Kitting = packing complementary products into one package sold as a "kit". S78
Seasonal storage	Hold seasonal or excess stock for peaks; avoid stock-outs and overstock. S79
Inventory management	Store until needed for production or distribution. S79
Strategic location	Near customers or transport hubs; lower cost, faster delivery. S79
Risk mitigation	Buffer against supplier delays, transport issues, natural disasters. S79

Cross docking



Original slide S80. Single-product trucks unload at the **inbound area**; the **cross docking terminal** does screening and sorting; mixed loads leave from the **outbound area**.

UTILITIES (5) S81	CHALLENGES (4) S82	IMPACTS (4) S83
Just-in-time (JIT) inventory	Supplier and customer relationships	Enhanced customer service
Optimized transportation (consolidates small shipments)	Quality control (no normal warehousing checks)	Improved efficiency
Reduced inventory holding costs	Coordination and synchronization of inbound and outbound	Cost savings
Improved efficiency	Technology and infrastructure (high setup cost)	Competitive advantage
Faster transit times (lower lead time)		

Case studies

Amazon	Fast-delivery retail model. Opened what it called its **first inbound cross-docking distribution centre in Germany**: receives from suppliers and distributes across Europe **within a day**; robots work alongside humans. S84
Roche Diagnostics (Switzerland)	Pharma; mostly **perishable** products needing **same-day** delivery. Uses cross-docking for **temperature control** and fast delivery. S85
Mainfreight (New Zealand)	**Offers** cross-docking services in NZ and worldwide: consolidate freight, cut warehousing cost, save time, fewer bottlenecks. Best-suited products: perishables, retail products with constant demand, pre-barcoded ready-for-sale goods. S86

Conclusion: effective storage management gives a smooth material flow, optimized inventory and met demand. The warehouse is for temporary storage at a strategic location and sometimes adds value. S87

LECTURE 05 S90–S127

Logistics Services Providers

Covers: LSP introduction, key components, outsourcing logistics, 1PL to 5PL, case study S91

> LOGISTICS SERVICE PROVIDER (LSP) · EXACT WORDS
>
> *"A logistics service provider (LSP) is a company or organization that offers a range of logistical services to support the transportation, storage, distribution, and management of goods and materials for businesses."* S92

Key components of LSPs 8 items

Services offered	Transport (road, rail, air, sea), warehousing, freight forwarding, customs brokerage, inventory, fulfilment, packaging, consulting, technology. S94
Customization	Tailored by industry, location, volume, handling needs, budget. S94
Expertise & experience	Specialists in transport, warehousing, customs compliance, inventory optimization. S95
Global reach	Global presence or partner networks. S95
Technology integration	**TMS, WMS**, GPS, RFID, cloud platforms. S96
Risk management	Insurance, contingency planning, regulatory compliance. S96
Cost efficiency	Economies of scale; pricing such as **pay-per-use** or **volume-based**. S97
Continuous improvement	Track metrics, gather feedback, invest in tech and process upgrades. S97

Outsourcing logistics: industry trends

- A.T. Kearney survey, why companies outsource (in the order shown, tallest bar first): **avoid capital expenditures**, cut costs, increase flexibility (service/financial), focus on core business, improve service, lack of scale, acquire talent, avoid labour problems. S99
- Most outsourced: **domestic transportation 83%**, warehousing 66%, international transportation 63%, customs brokerage 46%, freight forwarding 46%. Least: fleet management 10%, customer service 11%, LLP/4PL services 11%. S100
- Activities that are more **strategic and customer-facing** are outsourced less often. S100

1PL to 5PL at a glance

TYPE	WHO / WHAT	ASSET-BASED?	EGGS EXAMPLE S101
1PL	Firm or individual with its own cargo that transports it; the **consigner**. **Two parties** (supplier and retailer/customer), **no middlemen**. S102	Own assets	A farmer who delivers eggs to a grocery store to sell
2PL	**Asset-based carrier** that owns the means of transport (rail, road, sea, air). E.g. airlines, shipping lines, hauling companies. Used for international transport of heavy and wholesale goods. S105	Yes	A courier delivers eggs from farm to grocery store
3PL	Transport and delivery **plus** warehousing, terminal operations, customs brokerage, SCM, IT for tracking. S108	Usually	A fulfilment company with a fleet of trucks transports eggs
4PL	Overseer managing an **entire supply chain**; called **Lead Logistics Provider**; like a consultant; contracts many 3PLs; **single invoice**; no physical means to move inventory. S114	No	A company that manages a 3PL on the farmer's behalf
5PL	**Logistics aggregator**; bundles demand of 3PLs into bulk to get better rates from airlines and shipping companies; one step above a 4PL. S120	No	A company that manages the farmer's supply chain from production to delivery

1PL components

- **Ownership and control**: the company runs its own transport, warehousing, distribution and inventory. S103
- **In-house operations**: own fleet, warehouses, staff, tech. S103
- **Vertical integration** with manufacturing, procurement and sales. S103
- Benefits: control over quality, security, customer service. Challenges: costly fleets and warehouses, hard to scale. Risk: the company carries all of it. S104

2PL components

- Narrower **service scope** than 3PLs; may specialize in transport, warehousing, freight forwarding, customs brokerage or fulfilment. S106
- Specialized expertise (e.g. a transport 2PL with trucks, drivers and a TMS); customized solutions. S106
- Integration with client operations; scalability and flexibility without long-term commitments; technology (TMS, WMS, tracking). S107

3PL components

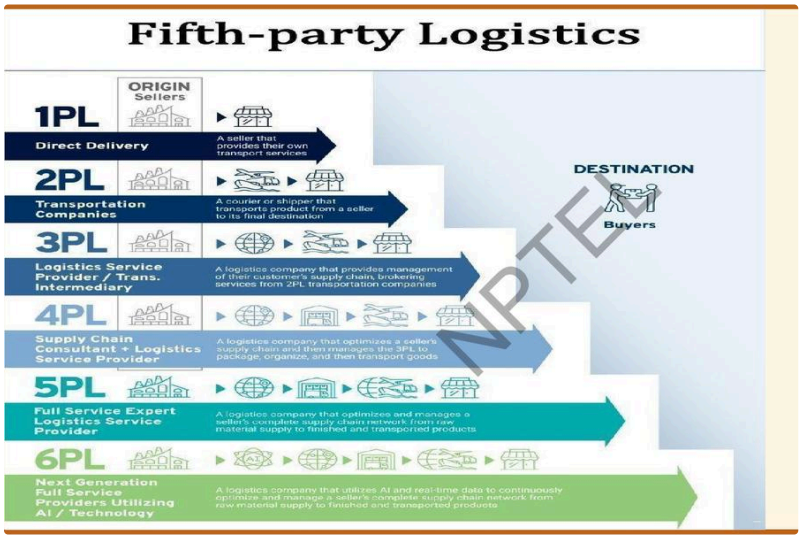
- 90% of Fortune 500 companies use a 3PL (Armstrong & Associates). Flow: Manufacturer → [packaging, warehousing, transportation] → Market. S109
- Services pie: transportation management **22%**, warehousing 21%, value added 19%, integrated 9%, international 9%, private fleets 7%, other 5%, lead logistics 4%, intermodal 4%. S108
- Example, Hitachi Transport System: procurement, production, sales and reverse logistics services, plus collaborative logistics within the same industry. S110
- Components: comprehensive services, outsourced expertise (focus on core competencies), scalability and flexibility S111; global reach, technology integration S112; cost savings, risk management S113.

4PL components

- **Integration and coordination** of multiple 3PLs and partners; the 4PL is the orchestrator. S116
- **End-to-end visibility**: consolidates data from 3PLs, ERP systems and IoT devices. S116
- Customized solutions; technology integration (TMS, WMS, analytics, cloud) S117; risk management; continuous improvement S118; global reach; strategic insights and planning S119.

5PL and beyond

- Not asset-based; works across all disciplines; a fully integrated solution covering the whole chain through multiple outsourced providers. S120



Original slide S122, "Fifth-party Logistics". 1PL direct delivery · 2PL transportation companies · 3PL LSP / transport intermediary · 4PL supply chain consultant + LSP · 5PL full-service expert LSP · **6PL** next-generation providers using AI / technology.

Case study: Elixia and a leading Indian 3PL

- Client: a leading Indian 3PL covering **1,00,000+ km/month**; inbound, outbound, primary and secondary distribution, last-mile delivery. S123
- Problem: **improper vehicle allocation**, vehicles running at **incomplete capacity**, **inefficient route planning** for local and upcountry deliveries. S124
- Result: clear budgeted costs and vehicle needs; INR 6.7 lac saved in one month by optimizing load, space and route. S124
- Headline figures: **8.4%** cost savings after route and space optimization · **650 Tn** carbon savings · **25%** increase in vehicle space utilization. S125

BEFORE THE EXAM ALL LECTURES

Cheat sheet

Numbers to remember

NUMBER	MEANING	SLIDE
≥ 50%	Transportation share of total supply chain costs (CBRE)	S3
58%	Transportation share of logistics costs by function	S3
4,132 bn	Trucking, the costliest mode (USD)	S3
8.4 tn €	Global logistics industry, 2021 (13.7 tn € by 2027)	S47
9 tn $	Global logistics costs 2020 = 10.7% of global GDP	S47
24.0%	Indonesia, highest logistics cost/GDP in 2016 chart (Korea lowest, 7.8%)	S46
$274 bn	Indian logistics market, 2022	S49
$563 bn	India 2030 projection, 9.4% CAGR	S49
14% vs 11%	India logistics cost/GDP vs BRICS average	S49
90%	Fortune 500 companies using a 3PL	S109
83%	Shippers outsourcing domestic transportation	S100
22%	Transportation management, biggest 3PL service slice	S108
₹6.7 lac	Elixia case: saved in one month	S124
8.4% · 650 Tn · 25%	Elixia: cost savings · carbon savings · vehicle space utilization	S125

Lists and their sizes

7 R's	product, quantity, time, condition, place, customer, (financial) resources S8
5 key elements	transportation, warehousing, inventory control, order fulfillment, distribution S10
7 advantages	efficiency, cost reduction, customer satisfaction, competitive advantage, risk management, resource optimization, market expansion S20
8 core elements	strategic planning, inventory, warehousing, transportation, order processing, distribution, technology, continuous improvement S24
8 modern concepts	lean, JIT, agile, green, collaborative, omni-channel, reverse, visibility S27
9 development factors	globalization, e-commerce, collaboration, technology, infrastructure, emerging markets, sustainability, customer expectations, regulation S39
12 Indian challenges	see the L3 table S51
FAST	Flow, Accessibility, Space, Throughput S75
U / I / L	most common / high-volume through-flow / heavy storage S76
7 warehouse utilities	consolidation, cross-docking, value-added services, seasonal storage, inventory management, strategic location, risk mitigation S78
5 · 4 · 4	cross-docking utilities · challenges · impacts S81
8 LSP components	services, customization, expertise, global reach, technology, risk, cost efficiency, continuous improvement S94

MEMORY HOOK (MY OWN, NOT FROM THE SLIDES)

Climb the PL ladder by how much of the chain is handed over. 1PL does it yourself. 2PL hires a truck. 3PL hires a truck company that also stores and packs. 4PL hires a manager who runs several 3PLs (one invoice, no trucks). 5PL pools many 3PLs' demand to bargain for bulk rates.

READ ONCE SOURCE CHECK

Slide quirks to watch

- **Euros or dollars?** The S45 chart shows 8.43 (2021) and 13.71 (2027) in *trillion US dollars*; S47 writes the same figures as "8.4 trillion euros" and "13.7 trillion euros". If a question quotes S47, answer in euros. S45 S47
- **India's cost/GDP:** 13.0% in the 2016 chart vs 14% in the text. When a question pairs it with the BRICS average (11%), use 14%. S46 S49
- **2PL scope:** S105 says 2PLs focus "solely on the transportation sector"; S106 says they may specialize in transport, warehousing, freight forwarding, customs brokerage or fulfilment. S105 S106
- **5PL picture:** the S121 5PL diagram is the same image as the S115 4PL diagram. Rely on the S120 text for 5PL. S115 S121
- **6PL** appears only in the S122 graphic, not in the lecture text. S122
- **"Right price"** appears in the S70 storage diagram but is *not* one of the Seven R's on S8. S8 S70
- Spellings as written: "A.T. Kearny" (title text) vs "A.T. Kearney" (source line); "Costal ports" means coastal. S99 S55
- Video or link-only slides (content not in the PDF): S4, S9, S38, S73, S98. S93 is blank apart from its title; S77 is a warehouse-layout picture with no text.

SELF-TEST 50 QUESTIONS

Week 1 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 1

According to Lecture 1, logistics manages the flow of goods, information and resources from:

- **A** the point of origin to the point of consumption
- **B** the port of loading to the port of destination
- **C** the factory gate to the retailer
- **D** the supplier to the warehouse

Q2 LECTURE 1

Which statement about the direction of flow in logistics matches the slides?

- **A** Flow of information only
- **B** Forward as well as reverse flow
- **C** Reverse flow only
- **D** Forward flow only

Q3 LECTURE 1

Which of the following is NOT one of the Seven R's of logistics?

- **A** Right condition
- **B** Right quantity
- **C** Right supplier
- **D** Right (financial) resources

Q4 LECTURE 1

In the Seven R's, 'right product' explicitly includes:

- **A** the right price for it
- **B** the right information about it
- **C** the right packaging
- **D** the right brand name

Q5 LECTURE 1

The logistics evolution slide shows logistics growing in which order?

- **A** Local → Regional → National → Global
- **B** Local → National → Regional → Global
- **C** Regional → Local → National → Global
- **D** Global → National → Regional → Local

Q6 LECTURE 1

In the 'Total Supply Chain Costs' chart (CBRE, 2014), transportation accounts for:

- **A** at least 50%
- **B** 15%
- **C** 20%
- **D** 10%

Q7 LECTURE 1

In the 'Logistics Costs by Function' pie chart, which function has the largest share?

- **A** Transportation
- **B** Warehousing
- **C** Inventory carrying
- **D** Administrative

Q8 LECTURE 1

By mode, which has the highest logistics cost on the slide (billions USD)?

- **A** Trucking
- **B** Air
- **C** Maritime
- **D** Rail

Q9 LECTURE 1

Which of these is NOT listed as a key element of logistics?

- **A** Inventory control
- **B** Order fulfillment
- **C** Distribution
- **D** Product advertising

Q10 LECTURE 1

The ultimate aim of logistics, as stated in Lecture 1, is to:

- **A** deliver the correct products to the right location at the appropriate time, while minimizing expenses and maximizing effectiveness
- **B** own as many vehicles as possible
- **C** hold maximum inventory at every node
- **D** replace suppliers with in-house production

Q11 LECTURE 2

Customers track their orders using:

- **A** warehouse bin codes
- **B** bank reference numbers
- **C** docket numbers supplied by the logistics service provider
- **D** GST numbers

Q12 LECTURE 2

Who may serve last-mile delivery, according to Lecture 2?

- **A** Port authorities
- **B** Customs brokers
- **C** Local carriers or postal services
- **D** Raw-material suppliers

Q13 LECTURE 2

Which advantage of effective logistics management 'enables companies to reach new customers in different geographic regions efficiently'?

- **A** Resource optimization
- **B** Market expansion
- **C** Risk management
- **D** Cost reduction

Q14 LECTURE 2

Mitigating stock-outs, transportation delays and supply chain disruptions to ensure business continuity is the advantage called:

- **A** Competitive advantage
- **B** Efficiency
- **C** Risk management
- **D** Customer satisfaction

Q15 LECTURE 2

In the logistics network diagram, what comes immediately after 'Plants'?

- **A** Customers
- **B** Port of Destination
- **C** Port of Loading
- **D** Warehouses

Q16 LECTURE 2

'Defining logistics objectives, analysing market demands, and aligning strategies with business goals' describes which core element?

- **A** Strategic planning
- **B** Continuous improvement
- **C** Order processing
- **D** Distribution

Q17 LECTURE 2

Which modern logistics concept draws on lean manufacturing and focuses on waste reduction?

- **A** Collaborative logistics
- **B** Agile logistics
- **C** Lean logistics
- **D** Omni-channel logistics

Q18 LECTURE 2

Just-in-Time (JIT) logistics mainly aims to minimize:

- **A** sales channels
- **B** packaging material
- **C** the number of customers
- **D** inventory holding costs

Q19 LECTURE 2

Green logistics is also known as:

- **A** Omni-channel logistics
- **B** Agile logistics
- **C** Reverse logistics
- **D** Sustainable logistics

Q20 LECTURE 2

Supply chain visibility relies on which technologies, per the slides?

- **A** Fax and telex
- **B** RFID, AI-ML, IoT, UAV, GPS and cloud-based software
- **C** Manual record books only
- **D** Barcode printers only

Q21 LECTURE 3

Which of these is NOT a task of manufacturing logistics as listed in Lecture 3?

- **A** Scheduling production
- **B** Distributing finished products
- **C** Setting retail prices
- **D** Procurement

Q22 LECTURE 3

'Streamlining material movement within the manufacturing facility through suitable equipment and processes' is which architecture component?

- **A** Transportation
- **B** Quality control
- **C** Material handling
- **D** Warehousing

Q23 LECTURE 3

Which of the following is NOT a factor contributing to the development of logistics?

- **A** E-commerce growth
- **B** Decline of international trade
- **C** Regulatory changes
- **D** Globalization

Q24 LECTURE 3

Per the slide text, the global logistics industry was worth over ____ in 2021.

- **A** 13.7 trillion euros
- **B** 85.24 trillion US dollars
- **C** 4.2 trillion euros
- **D** 8.4 trillion euros

Q25 LECTURE 3

Global total logistics costs in 2020 (about 9 trillion USD) were roughly what share of global GDP?

- **A** 8.4%
- **B** 10.7%
- **C** 14%
- **D** 5.2%

Q26 LECTURE 3

Which region has the largest share of logistics costs, and which country is the largest 3PL market?

- **A** North America; China
- **B** Asia-Pacific; United States
- **C** Asia-Pacific; India
- **D** Europe; Germany

Q27 LECTURE 3

The Indian logistics market (about $274 billion in 2022) is projected to reach about ____ by 2030 at a 9.4% CAGR.

- **A** $363 billion
- **B** $563 billion
- **C** $674 billion
- **D** $463 billion

Q28 LECTURE 3

India's logistics cost as a share of GDP is ____, against a BRICS average of ____.

- **A** 14%; 11%
- **B** 10.7%; 14%
- **C** 11%; 14%
- **D** 8.2%; 11%

Q29 LECTURE 3

Mass delivery delays at festival time are the slide's example of which challenge?

- **A** Ever-increasing fuel costs
- **B** Customer's mindset
- **C** Rail tariffs
- **D** High order intensity ratio

Q30 LECTURE 3

Why do large vessels sometimes fail to dock at nearer coastal ports in India?

- **A** Strict customs rules
- **B** High rail tariffs
- **C** Shortage of drivers
- **D** Lack of depth at the ports

Q31 LECTURE 4

Blinkit, Zepto and similar 10-minute deliveries depend on:

- **A** Ports of destination
- **B** Dark stores
- **C** 5PL aggregators
- **D** Bonded warehouses

Q32 LECTURE 4

A bonded warehouse is used for:

- **A** items under customs control
- **B** perishable goods
- **C** returned goods
- **D** immediate transfer without storage

Q33 LECTURE 4

AS/RS, listed as storage equipment, stands for:

- **A** Advanced Shipping and Routing Software
- **B** Automatic Sorting and Racking Shelves
- **C** Automated Storage and Retrieval Systems
- **D** Asset Storage and Return Service

Q34 LECTURE 4

In the FAST factors of warehouse layout design, 'T' stands for:

- **A** Tracking
- **B** Time
- **C** Throughput
- **D** Transport

Q35 LECTURE 4

Throughput in a warehouse typically quantifies:

- **A** empty pallet storage
- **B** inbound operations
- **C** office space
- **D** outbound operations

Q36 LECTURE 4

Which warehouse layout is described as the most common, working with any type of warehouse?

- **A** L shape
- **B** I shape
- **C** T shape
- **D** U shape

Q37 LECTURE 4

Warehouse 'kitting' means:

- **A** counting stock in cycles
- **B** packing complementary individual products into one package sold as a kit
- **C** holding seasonal stock
- **D** labelling pallets with RFID

Q38 LECTURE 4

Cross-docking directly supports which inventory practice?

- **A** Just-in-Time (JIT)
- **B** Seasonal storage
- **C** Long-term bonded storage
- **D** Building safety stock

Q39 LECTURE 4

Which of these is a CHALLENGE of cross-docking (not a utility or impact)?

- **A** Cost savings
- **B** Coordinating and synchronizing inbound and outbound shipments
- **C** Enhanced customer service
- **D** Faster transit times

Q40 LECTURE 4

Where did Amazon open what it called its first inbound cross-docking distribution centre?

- **A** New Zealand
- **B** United States
- **C** Germany
- **D** India

Q41 LECTURE 5

A first-party logistics (1PL) arrangement consists of:

- **A** a shipper, a carrier and a 3PL
- **B** a lead logistics provider managing many 3PLs
- **C** two parties (supplier and retailer/customer) with no middlemen
- **D** an aggregator bundling 3PL demand

Q42 LECTURE 5

1PL often involves ____, integrating logistics with manufacturing, procurement and sales.

- **A** vertical integration
- **B** horizontal outsourcing
- **C** cross-docking
- **D** freight aggregation

Q43 LECTURE 5

A 2PL is best described as:

- **A** a supply chain consultant
- **B** an asset-based carrier that owns the means of transportation
- **C** a non-asset logistics aggregator
- **D** a seller delivering its own goods

Q44 LECTURE 5

According to Armstrong & Associates, what share of Fortune 500 companies use a 3PL?

- **A** 50%
- **B** 100%
- **C** 90%
- **D** 75%

Q45 LECTURE 5

In the 'Services provided by 3PLs' pie chart, the largest slice is:

A International (9%)
B Value added (19%)
C Warehousing (21%)
D Transportation management (22%)

Q46 LECTURE 5

A 4PL is often called a:

A Logistics aggregator
B Lead Logistics Provider
C First-party consigner
D Asset-based carrier

Q47 LECTURE 5

A 5PL is called a logistics aggregator because it:

A handles only customs clearance
B delivers its own goods directly
C owns the largest truck fleet
D aggregates the demands of 3PLs into bulk volume to get better rates from airlines and shipping companies

Q48 LECTURE 5

In the eggs example, 'a logistics company that manages a 3PL on behalf of the farmer to package and deliver eggs to the grocery store' is a:

A 4PL
B 3PL
C 2PL
D 5PL

ANSWERS 50 QUESTIONS

Answer key

Q	ANS	WHY · SLIDE
Q1	A	The definition on S7 says 'from the point of origin to the point of consumption'. S7
Q2	B	S7: 'Forward as well as reverse flow.' S7
Q3	C	The Seven R's: product, quantity, time, condition, place, customer, (financial) resources. 'Supplier' is not one. S8
Q4	B	S8: 'Right product (including the right information about it)'. S8
Q5	A	S5 labels the stages Local, Regional, National, Global. S5
Q6	A	Transportation at least 50%, inventory 20%, labor 15%, other 10%, occupancy 5%. S3
Q7	A	Transportation 58%, inventory carrying 23%, warehousing 11%, administrative 8%. S3
Q8	A	Trucking 4,132 vs maritime 714, air 301, rail 297. S3
Q9	D	Key elements: transportation, warehousing, inventory control, order fulfillment, distribution. S10
Q10	A	Quoted directly from S10. S10
Q11	C	S18, Order Tracking: 'tracking respective docket numbers supplied by the logistics service provider'. S18
Q12	C	S18, Delivery: 'Local carriers or postal services could serve for last-mile delivery.' S18
Q13	B	That is the description of Market Expansion on S21. S21
Q14	C	S21: Risk Management. S21
Q15	C	Suppliers → Plants → Port of Loading → Port of Destination → Warehouses → Customers. S22
Q16	A	S24: Strategic Planning. S24
Q17	C	S27: Lean Logistics focuses on waste reduction and streamlining processes. S27
Q18	D	S27: JIT aligns production and delivery precisely with customer demand to minimize inventory holding costs. S27
Q19	D	S28: 'Also known as sustainable logistics'. S28
Q20	B	S29 lists RFID, AI-ML, IoT, UAV, GPS and cloud-based software solutions. S29
Q21	C	Tasks: procurement, inventory control, scheduling production, managing transportation, distributing finished products. S35
Q22	C	S37: Material Handling. S37
Q23	B	The nine factors are globalization, e-commerce growth, supply chain collaboration, technology, infrastructure, emerging markets, sustainability, customer expectations and regulatory changes. S39–S44
Q24	D	S47: over 8.4 trillion euros in 2021, projected to exceed 13.7 trillion euros by 2027. S47
Q25	B	S47: about 10.7% of global GDP (85.24 trillion USD). S47
Q26	B	S47: Asia-Pacific has the largest share of costs; the US is the largest 3PL market. S47
Q27	B	S49: $563 billion by 2030 at 9.4% CAGR. S49
Q28	A	S49: 14% for India vs 11% BRICS average. S49
Q29	D	S52: festival time is the best example under High Order Intensity Ratio. S52
Q30	D	S55: large vessels fail to dock due to lack of depth, so companies choose farther ports. S55
Q31	B	S69: 'Enter: Dark Stores (and importance of storage)'. S69
Q32	A	S71: bonded warehouses for items under customs control. Cold storage is for perishables; cross-docking for immediate transfer. S71
Q33	C	S72: automated storage and retrieval systems (AS/RS). S72
Q34	C	FAST = Flow, Accessibility, Space, Throughput. S75
Q35	D	S75: 'Throughput typically quantifies the outbound operations within a warehouse.' S75
Q36	D	U shape is the most common. I shape suits high volumes; L shape suits heavy storage needs. S76
Q37	B	S78 defines kitting as packing complementary products into a single package sold together as a 'kit'. S78
Q38	A	S81: cross-docking facilitates just-in-time (JIT) inventory practices. S81
Q39	B	S82 challenges: supplier and customer relationships, quality control, coordination and synchronization, technology and infrastructure. S82
Q40	C	S84: Germany; it distributes across Europe within a day. S84
Q41	C	S102: 1PL is made up of two parties, the supplier and the retailer/customer; no middlemen. S102
Q42	A	S103: Vertical Integration. S103
Q43	B	S105: examples are airlines, shipping lines and hauling companies. S105
Q44	C	S109: 90% of Fortune 500 companies use a 3PL. S109
Q45	D	Transportation management 22% is just ahead of warehousing 21%. S108
Q46	B	S114: 4PLs are often called Lead Logistics Providers, give a single-invoice solution and lack physical means to move inventory. S114
Q47	D	S120: 5PLs bundle the needs of multiple 3PLs to get better rates; not asset-based. S120
Q48	A	S101: that description is under 4PL. S101
Q49	A	Domestic transportation 83%, warehousing 66%, international transportation 63%. Fleet management is lowest at 10%. S100

Q	ANS	WHY · SLIDE
Q50	D	S124: INR 6.7 lac in one month. S125 adds 8.4% cost savings, 650 Tn carbon and 25% better vehicle space use. S124

Made from *WEEK 1.pdf* (127 slides, NPTEL Logistics & Supply Chain Management, Prof. Vikas Thakur, IIT Kharagpur). Notes use only the slides, except the box marked "my own".